CorroSight Evidence Passport

CorroSight by CorroGuard AI - pipeline inspection comparison, explainable review priorities and evidence trail.

This report was generated from FICTIONAL SYNTHETIC data for demonstration. Review priorities are recommendations for human review. They are not risk classifications, failure probabilities or repair decisions. No remaining-strength assessment is performed. Qualified engineering review is required before any action.

1. Inspection evidence

Item	Value
report	CorroSight Evidence Passport
generated utc	2026-10-23 08:52:02 UTC
software	CorroSight by CorroGuard AI v1.0.0-prototype
dataset	Demo dataset (synthetic, fictional pipeline CG-DEMO-24IN-01)
synthetic data	YES - fictional demonstration data
source files	synthetic_inspections.csv
dataset sha256	a0b6f5a021f4d3fd6979698500c4b94da3e447c99979c7b5c613f6294d060d27
analysis created utc	2026-10-23 08:51:58 UTC
runs compared	R2016 (2016-01-15) -> R2023 (2023-01-15), 7.00 yr
forecast method	baseline
analysis snapshot id	(sample - no snapshot saved)
pipeline id(s)	CG-DEMO-24IN-01
records (earlier / later run)	266 / 342
segments covered	40
findings (tracks)	367

2. Data validation findings

Errors: 12 Warnings: 8 Info: 1. Source records are never deleted or modified; problem records are flagged and excluded only from the calculation they would corrupt.

ID	Severity	Code	Run	Observation	Message	Effect on analysis	Suggested action
V001	Error	MISSING_POSITION	R2016	A16-0229	Axial position is missing.	Excluded from matching (cannot be located); record retained.	Supply the reported odometer distance.
V002	Error	INVALID_POSITION	R2023	B23-0124	Axial position 5230.40 m is outside the valid range (0 - 4800 m).	Excluded from matching; record retained.	Check odometer reference and units.
V003	Error	INVALID_WALL_THICKNESS	R2016	A16-0075	Nominal wall thickness 0.0 mm is zero, negative or implausible.	Treated as unavailable for this record.	Correct the wall thickness in the source file.
V004	Error	MISSING_DEPTH	R2016	A16-0181	Depth is missing.	Excluded from growth calculation; still eligible for location matching.	Request the sizing result from the inspection vendor.
V005	Error	MISSING_DEPTH	R2016	A16-0196	Depth is missing.	Excluded from growth calculation; still eligible for location matching.	Request the sizing result from the inspection vendor.
V006	Error	MISSING_DEPTH	R2023	B23-0007	Depth is missing.	Excluded from growth calculation; still eligible for location matching.	Request the sizing result from the inspection vendor.

ID	Severity	Code	Run	Observation	Message	Effect on analysis	Suggested action
V007	Error	MISSING_DEPTH	R2023	B23-0166	Depth is missing.	Excluded from growth calculation; still eligible for location matching.	Request the sizing result from the inspection vendor.
V008	Error	MISSING_DEPTH	R2023	B23-0255	Depth is missing.	Excluded from growth calculation; still eligible for location matching.	Request the sizing result from the inspection vendor.
V009	Error	MISSING_DEPTH	R2023	B23-0269	Depth is missing.	Excluded from growth calculation; still eligible for location matching.	Request the sizing result from the inspection vendor.
V010	Error	NON_POSITIVE_DEPTH	R2023	B23-0232	Depth is zero or negative.	Excluded from growth calculation; still eligible for location matching.	Check the sign convention and the vendor sizing record.
V011	Error	DEPTH_UNIT_SUSPECT	R2023	B23-0026	Depth 3260 mm is implausible; units may be micrometres or % wall thickness.	Excluded from growth calculation; value NOT rescaled.	Confirm the unit with the vendor and correct the source file.
V012	Error	DEPTH_EXCEEDS_WT	R2023	B23-0072	Depth 12.76 mm exceeds nominal wall thickness 11.1 mm.	Excluded from growth calculation; flagged for review (possible sizing or data-entry error).	Verify the measurement and the wall-thickness record.
V013	Warning	INVALID_CLOCK	R2023	B23-0239	Clock position is missing or outside 0-12 h.	Matching uses axial position only for this record (clock treated as unknown).	Check the clock-position convention (hours vs degrees).
V014	Warning	MISSING_WALL_THICKNESS	-	-	22 records have no nominal wall thickness (segments: S17).	Depth as % of wall thickness, screening-threshold projections and engineering readiness cannot be completed for these records. They are NOT lowered in review priority.	Request the nominal wall thickness from pipe-book / design records.
V015	Warning	EXACT_DUPLICATE	R2016	A16-0073	CSV line 75 is an exact copy of line 74.	Second copy excluded from matching; both rows retained in the source data and report.	Remove the duplicate from the source export if confirmed.
V016	Warning	POSSIBLE_DUPLICATE	R2023	B23-D01	B23-D01 is within 0.012 m, 0.00 h and 0.00 mm depth of B23-0078 in the same run (similar length).	Excluded from matching pending review; NOT deleted. Appears in the findings list.	Confirm with the vendor whether this is a double call or two separate anomalies.
V017	Warning	POSSIBLE_DUPLICATE	R2023	B23-D03	B23-D03 is within 0.012 m, 0.00 h and 0.00 mm depth of B23-0219 in the same run (similar length).	Excluded from matching pending review; NOT deleted. Appears in the findings list.	Confirm with the vendor whether this is a double call or two separate anomalies.
V018	Warning	POSSIBLE_DUPLICATE	R2023	B23-D02	B23-D02 is within 0.012 m, 0.00 h and 0.00 mm depth of B23-0263 in the same run (similar length).	Excluded from matching pending review; NOT deleted. Appears in the findings list.	Confirm with the vendor whether this is a double call or two separate anomalies.
V019	Warning	POSSIBLE_DUPLICATE	R2023	B23-D04	B23-D04 is within 0.012 m, 0.00 h and 0.00 mm depth of B23-0319 in the same run (similar length).	Excluded from matching pending review; NOT deleted. Appears in the findings list.	Confirm with the vendor whether this is a double call or two separate anomalies.
V020	Warning	POSSIBLE_DUPLICATE	R2023	B23-D05	B23-D05 is within 0.012 m, 0.00 h and 0.00 mm depth of B23-0337 in the same run (similar length).	Excluded from matching pending review; NOT deleted. Appears in the findings list.	Confirm with the vendor whether this is a double call or two separate anomalies.
V021	Info	MISSING_UNCERTAINTY	-	-	14 records have no stated depth measurement uncertainty.	No uncertainty interval or noise test is computed for changes involving these records; they are labelled 'uncertainty unknown' (not treated as certain).	Obtain the tool sizing specification from the inspection vendor.

Log: 6 record(s) flagged as duplicates and excluded from matching (retained and reported).

3. Corrosion Memory Graph - matching decisions

Match status	Count	Meaning
Likely match	212	One-to-one pairing with no competing candidate within the ambiguity margin.
Ambiguous match	29	Provisional pairing; another candidate fits almost as well. Not used for forecasts.
Not found in later run	23	Earlier record with no later partner.
New in later run	95	Later record with no earlier partner (new, previously undetected, or false call).
Excluded from matching	8	Duplicate, missing identifier or invalid location; record retained.

Method: per-segment odometer alignment (offset range +0.25 to +0.99 m), spatial gates |dx| <= 1.0 m and |dclock| <= 1.0 h, normalised distance cost (scales 0.25 m, 0.4 h, 20.0 mm), one-to-one Hungarian assignment, ambiguity margin 1.0. Depth is not used for matching. The agreement index exp(-d^2/2) is a similarity measure, not a probability.

Evaluation against synthetic ground truth (synthetic only - not real-world accuracy)

Metric	Value
True cross-run pairs in data	243
Likely-match precision	1.000 (212/212)
Ambiguous-pair precision	1.000 (29/29)
Recall (likely + ambiguous)	0.992
Recall (likely only)	0.872
False matches / missed true pairs	0 / 2

4. Growth estimation, forecast and uncertainty

Change vs measurement noise	Count
Change within measurement noise	181
Not computed	131
Increase exceeds measurement noise	42
Measurement uncertainty unknown	12
Decrease exceeds measurement noise (anomalous)	1

Forecast status	Count
Reliable forecast cannot be established	172
Upper screening scenario only (growth not distinguishable from noise)	158
Projection available (central + upper scenario)	37

Observed change = later depth - earlier depth (mm); annualised change = change / elapsed years. Negative changes are kept as observed. Measurement-noise sigma = sqrt(u1^2 + u2^2) using stated 1-sigma depth uncertainties; the 2-sigma interval covers measurement noise only (not matching, sizing bias or model form). Projections assume a constant rate over a 5-year horizon and are illustrative. Forecast method selected: baseline. Model B shrinkage fit: population mean 0.0675 mm/yr, between-defect variance 0.00848 (n=195).

Rate estimator (synthetic ground truth, n=195)	MAE mm/yr	RMSE mm/yr	Bias mm/yr	2-sigma interval coverage
Baseline: per-defect observed rate	0.0573	0.0714	-0.0041	0.969
Model B: empirical-Bayes shrinkage	0.0418	0.0536	-0.0041	0.954

Synthetic measurement noise was generated with exactly the stated uncertainty, so interval coverage here is optimistic by construction. These metrics do not establish real-world accuracy.

5. Why-Inspect-First review priorities

Category counts: Review first: 33; Verify match / data: 34; Additional data needed: 29; Monitor: 15; Routine review: 256. Category = the higher of the condition-based and evidence-based categories, so missing or uncertain evidence can only raise a finding. Thresholds are illustrative demo assumptions chosen to make the review rules explicit. They are not industry standards, code limits or validated engineering criteria.

Demo thresholds: depth review 40% WT; rate review 0.2 mm/yr (beyond 2-sigma noise); screening threshold 50% WT within 5 yr (upper scenario).

#	Finding	Category	Condition	Evidence	Match	Obs (earlier -> later)	Latest %WT	Rate mm/yr	System proposed next step	Review status
1	T0153	Review first	High	Limited	Likely match	A16-0122 -> B23-0140	-	0.329	Send for engineering review	Not reviewed
2	T0177	Review first	High	Adequate	Likely match	A16-0140 -> B23-0163	60.5	0.513	Send for engineering review	Not reviewed
3	T0158	Review first	High	Adequate	Likely match	A16-0127 -> B23-0145	50.4	0.589	Send for engineering review	Not reviewed
4	T0160	Review first	High	Limited	Likely match	A16-0129 -> B23-0147	87.5	0.501	Send for engineering review	Not reviewed
5	T0157	Review first	High	Adequate	Likely match	A16-0126 -> B23-0144	51.3	0.464	Send for engineering review	Not reviewed
6	T0291	Review first	High	Adequate	Likely match	A16-0219 -> B23-0270	55.6	0.283	Send for engineering review	Not reviewed
7	T0342	Review first	High	Adequate	Likely match	A16-0254 -> B23-0317	43.2	0.356	Send for engineering review	Not reviewed
8	T0352	Review first	High	Adequate	Likely match	A16-0258 -> B23-0325	48.9	0.163	Send for engineering review	Not reviewed
9	T0081	Review first	High	Adequate	Likely match	A16-0067 -> B23-0077	48.5	0.127	Send for engineering review	Not reviewed
10	T0113	Review first	High	Adequate	Likely match	A16-0087 -> B23-0102	42.5	0.366	Send for engineering review	Not reviewed
11	T0162	Review first	High	Limited	Likely match	A16-0131 -> B23-0149	61.7	0.353	Send for engineering review	Not reviewed
12	T0094	Review first	High	Adequate	Likely match	A16-0075 -> B23-0086	40.1	0.260	Send for engineering review	Not reviewed
13	T0307	Review first	High	Adequate	Likely match	A16-0230 -> B23-0284	45.7	0.101	Send for engineering review	Not reviewed
14	T0003	Review first	High	Adequate	Likely match	A16-0003 -> B23-0003	34.9	0.238	Send for engineering review	Not reviewed
15	T0207	Review first	High	Adequate	Likely match	A16-0162 -> B23-0190	39.1	0.296	Send for engineering review	Not reviewed
16	T0066	Review first	High	Adequate	Likely match	A16-0057 -> B23-0063	35.3	0.353	Send for engineering review	Not reviewed
17	T0142	Review first	High	Adequate	Likely match	A16-0113 -> B23-0129	38.8	0.241	Send for engineering review	Not reviewed
18	T0252	Review first	High	Adequate	Likely match	A16-0193 -> B23-0233	42.6	0.099	Send for engineering review	Not reviewed
19	T0337	Review first	High	Adequate	Likely match	A16-0251 -> B23-0312	42.5	0.067	Send for engineering review	Not reviewed
20	T0076	Review first	High	Adequate	Likely match	A16-0063 -> B23-0073	41.0	0.124	Send for engineering review	Not reviewed
21	T0021	Review first	High	Adequate	Likely match	A16-0017 -> B23-0020	41.4	0.107	Send for engineering review	Not reviewed
22	T0186	Review first	High	Adequate	Likely match	A16-0146 -> B23-0171	30.0	0.340	Send for engineering review	Not reviewed
23	T0134	Review first	High	Adequate	Likely match	A16-0107 -> B23-0121	33.6	0.187	Send for engineering review	Not reviewed
24	T0018	Review first	High	Adequate	Likely match	A16-0014 -> B23-0017	27.7	0.346	Send for engineering review	Not reviewed
25	T0073	Review first	High	Adequate	Likely match	A16-0061 -> B23-0070	40.6	0.020	Send for engineering review	Not reviewed
26	T0127	Review first	High	Adequate	Likely match	A16-0101 -> B23-0115	22.4	0.259	Send for engineering review	Not reviewed
27	T0356	Review first	High	Limited	Likely match	A16-0260 -> B23-0329	41.9	0.051	Send for engineering review	Not reviewed
28	T0143	Review first	High	Adequate	Likely match	A16-0114 -> B23-0130	24.7	0.210	Send for engineering review	Not reviewed
29	T0278	Review first	High	Adequate	Likely match	A16-0211 -> B23-0258	18.9	0.250	Send for engineering review	Not reviewed
30	T0203	Review first	High	Adequate	Likely match	A16-0158 -> B23-0185	18.3	0.243	Send for engineering review	Not reviewed
31	T0184	Review first	High	Adequate	Likely match	A16-0144 -> B23-0169	17.7	0.233	Send for engineering review	Not reviewed
32	T0163	Review first	High	Limited	Ambiguous match	A16-0132 -> B23-0151	27.2	0.210	Send for engineering review	Not reviewed

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33	T0221	Review first	High	Limited	Ambiguous match	A16-0174 -> B23-0203	24.8	0.201	Send for engineering review	Not reviewed
34	T0152	Verify match / data	Elevated	Limited	Ambiguous match	A16-0121 -> B23-0139	-	0.196	Verify the defect match / sizing against source ILI data	Not reviewed
35	T0151	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0120 -> B23-0138	-	-0.133	Verify the defect match / sizing against source ILI data	Not reviewed
36	T0082	Verify match / data	No rule triggered	Insufficient	Excluded from matching	- -> B23-D01	37.5	-	Verify the defect match / sizing against source ILI data	Not reviewed
37	T0345	Verify match / data	No rule triggered	Insufficient	Excluded from matching	- -> B23-D04	34.6	-	Verify the defect match / sizing against source ILI data	Not reviewed
38	T0104	Verify match / data	No rule triggered	Limited	Not found in later run	A16-0079 -> -	34.2	-	Verify the defect match / sizing against source ILI data	Not reviewed
39	T0097	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0077 -> B23-0089	30.3	0.113	Verify the defect match / sizing against source ILI data	Not reviewed
40	T0096	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0076 -> B23-0088	27.2	0.076	Verify the defect match / sizing against source ILI data	Not reviewed
41	T0266	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0205 -> B23-0246	26.7	0.066	Verify the defect match / sizing against source ILI data	Not reviewed
42	T0265	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0204 -> B23-0245	26.6	0.016	Verify the defect match / sizing against source ILI data	Not reviewed
43	T0173	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0138 -> B23-0160	26.5	0.023	Verify the defect match / sizing against source ILI data	Not reviewed
44	T0172	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0137 -> B23-0159	25.4	0.059	Verify the defect match / sizing against source ILI data	Not reviewed
45	T0239	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0183 -> B23-0220	24.9	0.029	Verify the defect match / sizing against source ILI data	Not reviewed
46	T0365	Verify match / data	No rule triggered	Insufficient	Excluded from matching	- -> B23-D05	22.4	-	Verify the defect match / sizing against source ILI data	Not reviewed
47	T0238	Verify match / data	No rule triggered	Insufficient	Excluded from matching	- -> B23-D03	21.7	-	Verify the defect match / sizing against source ILI data	Not reviewed
48	T0264	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0203 -> B23-0244	18.4	-0.027	Verify the defect match / sizing against source ILI data	Not reviewed
49	T0273	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0209 -> B23-0253	18.2	-0.084	Verify the defect match / sizing against source ILI data	Not reviewed
50	T0220	Verify match / data	Elevated	Limited	Ambiguous match	A16-0173 -> B23-0202	16.6	0.187	Verify the defect match / sizing against source ILI data	Not reviewed
51	T0263	Verify match / data	Elevated	Limited	Ambiguous match	A16-0202 -> B23-0243	15.2	0.144	Verify the defect match / sizing against source ILI data	Not reviewed
52	T0057	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0049 -> B23-0055	14.7	0.041	Verify the defect match / sizing against source ILI data	Not reviewed
53	T0274	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0210 -> B23-0254	13.2	0.011	Verify the defect match / sizing against source ILI data	Not reviewed
54	T0325	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0243 -> B23-0300	12.2	-0.139	Verify the defect match / sizing against source ILI data	Not reviewed
55	T0041	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0035 -> B23-0040	12.1	0.039	Verify the defect match / sizing against source ILI data	Not reviewed

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56	T0039	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0033 -> B23-0038	11.7	0.106	Verify the defect match / sizing against source ILI data	Not reviewed
57	T0038	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0032 -> B23-0037	11.4	0.049	Verify the defect match / sizing against source ILI data	Not reviewed
58	T0091	Verify match / data	No rule triggered	Insufficient	Excluded from matching	A16-0073 -> -	11.2	-	Verify the defect match / sizing against source ILI data	Not reviewed
59	T0120	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0094 -> B23-0108	11.1	0.057	Verify the defect match / sizing against source ILI data	Not reviewed
60	T0250	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0192 -> B23-0231	10.1	0.084	Verify the defect match / sizing against source ILI data	Not reviewed
61	T0065	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0056 -> B23-0062	9.7	0.063	Verify the defect match / sizing against source ILI data	Not reviewed
62	T0261	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0200 -> B23-0241	8.9	-0.081	Verify the defect match / sizing against source ILI data	Not reviewed
63	T0324	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0242 -> B23-0299	8.1	-0.034	Verify the defect match / sizing against source ILI data	Not reviewed
64	T0323	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0241 -> B23-0298	7.8	0.066	Verify the defect match / sizing against source ILI data	Not reviewed
65	T0260	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0199 -> B23-0240	7.8	-0.003	Verify the defect match / sizing against source ILI data	Not reviewed
66	T0284	Verify match / data	No rule triggered	Insufficient	Excluded from matching	- -> B23-D02	6.1	-	Verify the defect match / sizing against source ILI data	Not reviewed
67	T0040	Verify match / data	No rule triggered	Limited	Ambiguous match	A16-0034 -> B23-0039	4.2	-0.177	Verify the defect match / sizing against source ILI data	Not reviewed
68	T0148	Additional data needed	Elevated	Limited	Likely match	A16-0119 -> B23-0135	-	0.190	Request missing inputs: nominal wall thickness	Not reviewed
69	T0145	Additional data needed	Elevated	Limited	Likely match	A16-0116 -> B23-0132	-	0.163	Request missing inputs: nominal wall thickness	Not reviewed
70	T0154	Additional data needed	No rule triggered	Limited	Likely match	A16-0123 -> B23-0141	-	0.120	Request missing inputs: nominal wall thickness	Not reviewed
71	T0155	Additional data needed	No rule triggered	Limited	Likely match	A16-0124 -> B23-0142	-	0.070	Request missing inputs: nominal wall thickness	Not reviewed
72	T0146	Additional data needed	No rule triggered	Limited	Likely match	A16-0117 -> B23-0133	-	0.014	Request missing inputs: nominal wall thickness	Not reviewed
73	T0147	Additional data needed	No rule triggered	Limited	Likely match	A16-0118 -> B23-0134	-	-0.013	Request missing inputs: nominal wall thickness	Not reviewed
74	T0156	Additional data needed	No rule triggered	Limited	Likely match	A16-0125 -> B23-0143	-	-0.044	Request missing inputs: nominal wall thickness	Not reviewed
75	T0075	Additional data needed	Not assessable	Insufficient	New in later run	- -> B23-0072	-	-	Request missing inputs: valid depth	Not reviewed
76	T0149	Additional data needed	No rule triggered	Limited	New in later run	- -> B23-0136	-	-	Request missing inputs: nominal wall thickness	Not reviewed
77	T0150	Additional data needed	No rule triggered	Limited	New in later run	- -> B23-0137	-	-	Request missing inputs: nominal wall thickness	Not reviewed
78	T0181	Additional data needed	Not assessable	Insufficient	New in later run	- -> B23-0166	-	-	Request missing inputs: valid depth	Not reviewed

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79	T0251	Additional data needed	Not assessable	Insufficient	New in later run	- -> B23-0232	-	-	Request missing inputs: valid depth	Not reviewed
80	T0275	Additional data needed	Not assessable	Insufficient	New in later run	- -> B23-0255	-	-	Request missing inputs: valid depth	Not reviewed
81	T0366	Additional data needed	No rule triggered	Insufficient	Excluded from matching	- -> B23-0124	29.2	-	Request missing inputs: valid location / identifier	Not reviewed
82	T0027	Additional data needed	No rule triggered	Limited	Likely match	A16-0021 -> B23-0026	28.6	-	Request missing inputs: valid depth in both runs	Not reviewed
83	T0072	Additional data needed	No rule triggered	Limited	Likely match	A16-0060 -> B23-0069	27.7	0.303	Request missing inputs: depth measurement uncertainty	Not reviewed
84	T0313	Additional data needed	No rule triggered	Limited	Likely match	A16-0233 -> B23-0289	26.5	0.134	Request missing inputs: depth measurement uncertainty	Not reviewed
85	T0185	Additional data needed	No rule triggered	Limited	Likely match	A16-0145 -> B23-0170	19.8	0.203	Request missing inputs: depth measurement uncertainty	Not reviewed
86	T0013	Additional data needed	No rule triggered	Limited	Likely match	A16-0010 -> B23-0012	19.6	0.076	Request missing inputs: depth measurement uncertainty	Not reviewed
87	T0245	Additional data needed	No rule triggered	Limited	Likely match	A16-0189 -> B23-0226	18.9	-0.031	Request missing inputs: depth measurement uncertainty	Not reviewed
88	T0128	Additional data needed	No rule triggered	Limited	Likely match	A16-0102 -> B23-0116	16.0	0.174	Request missing inputs: depth measurement uncertainty	Not reviewed
89	T0108	Additional data needed	No rule triggered	Limited	Likely match	A16-0082 -> B23-0098	15.4	0.071	Request missing inputs: depth measurement uncertainty	Not reviewed
90	T0017	Additional data needed	No rule triggered	Limited	Likely match	A16-0013 -> B23-0016	14.8	-0.013	Request missing inputs: depth measurement uncertainty	Not reviewed
91	T0255	Additional data needed	No rule triggered	Limited	Likely match	A16-0196 -> B23-0235	11.3	-	Request missing inputs: valid depth in both runs	Not reviewed
92	T0311	Additional data needed	No rule triggered	Limited	Likely match	A16-0231 -> B23-0288	10.1	0.027	Request missing inputs: depth measurement uncertainty	Not reviewed
93	T0008	Additional data needed	No rule triggered	Limited	Likely match	A16-0008 -> B23-0007	8.9	-	Request missing inputs: valid depth in both runs	Not reviewed
94	T0290	Additional data needed	No rule triggered	Limited	Likely match	A16-0218 -> B23-0269	7.9	-	Request missing inputs: valid depth in both runs	Not reviewed
95	T0236	Additional data needed	No rule triggered	Limited	Likely match	A16-0181 -> B23-0218	7.4	-	Request missing inputs: valid depth in both runs	Not reviewed
96	T0367	Additional data needed	No rule triggered	Insufficient	Excluded from matching	A16-0229 -> -	0.6	-	Request missing inputs: valid location / identifier	Not reviewed
97	T0111	Monitor	Elevated	Adequate	Likely match	A16-0085 -> B23-0100	29.4	0.193	Include in next scheduled data review	Not reviewed
98	T0161	Monitor	Elevated	Adequate	Likely match	A16-0130 -> B23-0148	26.5	0.180	Include in next scheduled data review	Not reviewed
99	T0121	Monitor	Elevated	Adequate	Likely match	A16-0095 -> B23-0109	29.6	0.157	Include in next scheduled data review	Not reviewed
100	T0086	Monitor	Elevated	Adequate	Likely match	A16-0071 -> B23-0081	27.4	0.147	Include in next scheduled data review	Not reviewed
101	T0228	Monitor	Elevated	Adequate	Likely match	A16-0176 -> B23-0210	28.5	0.163	Include in next scheduled data review	Not reviewed
102	T0257	Monitor	Elevated	Adequate	Likely match	A16-0198 -> B23-0237	26.9	0.184	Include in next scheduled data review	Not reviewed
103	T0054	Monitor	Elevated	Adequate	Likely match	A16-0046 -> B23-0052	21.6	0.169	Include in next scheduled data review	Not reviewed
104	T0215	Monitor	Elevated	Adequate	Likely match	A16-0168 -> B23-0197	24.3	0.179	Include in next scheduled data review	Not reviewed

#	Finding	Category	Condition	Evidence	Match	Obs (earlier -> later)	Latest %WT	Rate mm/yr	System proposed next step	Review status
105	T0123	Monitor	Elevated	Adequate	Likely match	A16-0097 -> B23-0111	23.2	0.156	Include in next scheduled data review	Not reviewed
106	T0115	Monitor	Elevated	Adequate	Likely match	A16-0089 -> B23-0104	22.9	0.156	Include in next scheduled data review	Not reviewed
107	T0281	Monitor	Elevated	Adequate	Likely match	A16-0214 -> B23-0261	19.7	0.153	Include in next scheduled data review	Not reviewed
108	T0188	Monitor	Elevated	Adequate	Likely match	A16-0147 -> B23-0173	16.3	0.196	Include in next scheduled data review	Not reviewed
109	T0035	Monitor	Elevated	Adequate	Likely match	A16-0029 -> B23-0034	17.8	0.150	Include in next scheduled data review	Not reviewed
110	T0106	Monitor	Elevated	Limited	New in later run	- -> B23-0096	29.6	-	Include in next scheduled data review	Not reviewed
111	T0321	Monitor	Elevated	Adequate	Likely match	A16-0239 -> B23-0297	14.9	0.171	Include in next scheduled data review	Not reviewed

256 findings in 'Routine review' are listed in the CSV export.

6. Finding evidence sheets (10)

T0153 - Review first - Likely match

Evidence	Earlier (R2016, 2016-01-15)	Later (R2023, 2023-01-15)
Observation ID (source record)	A16-0122	B23-0140
OBSERVED axial position m	2009.970	2010.444
OBSERVED clock h	2.23	2.36
OBSERVED depth mm (1-sigma)	2.42 (0.40)	4.72 (0.30)
OBSERVED length / width mm	22 / 35	14 / 46
Nominal WT mm (design)	-	-
Record quality flags	MISSING_WALL_THICKNESS	MISSING_WALL_THICKNESS

Result	Value
CALCULATED observed change	2.30 mm over 7.00 yr
CALCULATED annualised change	0.329 mm/yr
UNCERTAINTY (measurement only)	sigma 0.071 mm/yr; 2-sigma range 0.186 to 0.471
Change vs noise	Increase exceeds measurement noise
Latest depth % WT	-
PREDICTED (illustrative)	Projection available (central + upper scenario): central 6.36 mm, upper 7.08 mm at +5 yr
Years to demo threshold (upper)	-
Forecast limitation	Constant-rate projection using the baseline observed rate; upper scenario adds 2 sigma of measurement noise. Only two observations: long-term trend is NOT established.

Basis	Detail
Match rationale	Aligned axial difference +0.03 m (raw +0.47 m, odometer offset +0.45 m), clock difference 0.13 h, length difference -9 mm. Normalised distance d=0.56 (gate 4.0). No competing candidate within the gate.
Condition indicators	C3: observed change +2.30 mm over 7.0 yr = +0.329 mm/yr (A16-0122 -> B23-0140) exceeds 2 sigma noise (0.50 mm) and >= demo rate 0.2 mm/yr.
Evidence quality	Limited: E8: nominal wall thickness missing; % WT and threshold scenarios cannot be computed.
Engineering readiness	Not ready - inputs missing. Missing: Nominal wall thickness. Verify: Tool sizing accuracy and any bias between inspection tools (always required)
System proposed next step	Send for engineering review
Human review	Not reviewed
Anomaly flags	Change large relative to uncertainty
Contradictions	none
Evidence gaps	Missing engineering input
Identity resolution (reviewer)	none recorded

T0177 - Review first - Likely match

Evidence	Earlier (R2016, 2016-01-15)	Later (R2023, 2023-01-15)
Observation ID (source record)	A16-0140	B23-0163
OBSERVED axial position m	2294.468	2294.936
OBSERVED clock h	7.81	7.69
OBSERVED depth mm (1-sigma)	3.13 (0.40)	6.72 (0.30)
OBSERVED length / width mm	86 / 55	87 / 63
Nominal WT mm (design)	11.10	11.10
Record quality flags	-	-

Result	Value
CALCULATED observed change	3.59 mm over 7.00 yr
CALCULATED annualised change	0.513 mm/yr
UNCERTAINTY (measurement only)	sigma 0.071 mm/yr; 2-sigma range 0.370 to 0.656
Change vs noise	Increase exceeds measurement noise
Latest depth % WT	60.5
PREDICTED (illustrative)	Projection available (central + upper scenario): central 9.28 mm, upper 10.00 mm at +5 yr
Years to demo threshold (upper)	0.0
Forecast limitation	Constant-rate projection using the baseline observed rate; upper scenario adds 2 sigma of measurement noise. Only two observations: long-term trend is NOT established.

Basis	Detail
Match rationale	Aligned axial difference -0.05 m (raw +0.47 m, odometer offset +0.52 m), clock difference 0.12 h, length difference +1 mm. Normalised distance d=0.36 (gate 4.0). No competing candidate within the gate.
Condition indicators	C1: latest depth 6.72 mm = 61% of nominal WT 11.1 mm (B23-0163, R2023) >= demo threshold 40%. | C2: upper screening scenario (0.656 mm/yr) reaches the demo 50% WT threshold in 0.0 yr (<= 5-yr horizon). Illustrative, not a deadline. | C3: observed change +3.59 mm over 7.0 yr = +0.513 mm/yr (A16-0140 -> B23-0163) exceeds 2 sigma noise (0.50 mm) and >= demo rate 0.2 mm/yr.
Evidence quality	Adequate: No evidence issue detected
Engineering readiness	Inputs present for an engineer-selected assessment (not performed here). Missing: none. Verify: Tool sizing accuracy and any bias between inspection tools (always required)
System proposed next step	Send for engineering review
Human review	Not reviewed
Anomaly flags	Unusual rate vs. population; Change large relative to uncertainty
Contradictions	none
Evidence gaps	none
Identity resolution (reviewer)	none recorded

T0158 - Review first - Likely match

Evidence	Earlier (R2016, 2016-01-15)	Later (R2023, 2023-01-15)
Observation ID (source record)	A16-0127	B23-0145
OBSERVED axial position m	2052.294	2052.502
OBSERVED clock h	8.94	8.80
OBSERVED depth mm (1-sigma)	0.67 (0.40)	4.79 (0.30)
OBSERVED length / width mm	40 / 40	40 / 43
Nominal WT mm (design)	9.50	9.50
Record quality flags	-	-

Result	Value
CALCULATED observed change	4.12 mm over 7.00 yr
CALCULATED annualised change	0.589 mm/yr
UNCERTAINTY (measurement only)	sigma 0.071 mm/yr; 2-sigma range 0.446 to 0.731
Change vs noise	Increase exceeds measurement noise
Latest depth % WT	50.4
PREDICTED (illustrative)	Projection available (central + upper scenario): central 7.73 mm, upper 8.45 mm at +5 yr
Years to demo threshold (upper)	0.0
Forecast limitation	Constant-rate projection using the baseline observed rate; upper scenario adds 2 sigma of measurement noise. Only two observations: long-term trend is NOT established.

Basis	Detail
Match rationale	Aligned axial difference -0.20 m (raw +0.21 m, odometer offset +0.41 m), clock difference 0.14 h, length difference +0 mm. Normalised distance d=0.88 (gate 4.0). No competing candidate within the gate.
Condition indicators	C1: latest depth 4.79 mm = 50% of nominal WT 9.5 mm (B23-0145, R2023) >= demo threshold 40%. | C2: upper screening scenario (0.731 mm/yr) reaches the demo 50% WT threshold in 0.0 yr (<= 5-yr horizon). Illustrative, not a deadline. | C3: observed change +4.12 mm over 7.0 yr = +0.589 mm/yr (A16-0127 -> B23-0145) exceeds 2 sigma noise (0.50 mm) and >= demo rate 0.2 mm/yr.
Evidence quality	Adequate: No evidence issue detected
Engineering readiness	Inputs present for an engineer-selected assessment (not performed here). Missing: none. Verify: Tool sizing accuracy and any bias between inspection tools (always required)
System proposed next step	Send for engineering review
Human review	Not reviewed
Anomaly flags	Unusual rate vs. population; Change large relative to uncertainty
Contradictions	none
Evidence gaps	none
Identity resolution (reviewer)	none recorded

T0160 - Review first - Likely match

Evidence	Earlier (R2016, 2016-01-15)	Later (R2023, 2023-01-15)
Observation ID (source record)	A16-0129	B23-0147
OBSERVED axial position m	2094.259	2094.683
OBSERVED clock h	10.42	10.74
OBSERVED depth mm (1-sigma)	4.80 (-)	8.31 (0.30)
OBSERVED length / width mm	65 / 7	66 / 16
Nominal WT mm (design)	9.50	9.50
Record quality flags	MISSING_UNCERTAINTY	-

Result	Value
CALCULATED observed change	3.51 mm over 7.00 yr
CALCULATED annualised change	0.501 mm/yr
UNCERTAINTY (measurement only)	sigma - mm/yr; 2-sigma range - to -
Change vs noise	Measurement uncertainty unknown
Latest depth % WT	87.5
PREDICTED (illustrative)	Reliable forecast cannot be established: central - mm, upper - mm at +5 yr
Years to demo threshold (upper)	0.0
Forecast limitation	Measurement uncertainty unknown: the observed change cannot be distinguished from measurement noise, so no forecast is made.

Basis	Detail
Match rationale	Aligned axial difference +0.01 m (raw +0.42 m, odometer offset +0.41 m), clock difference 0.32 h, length difference +1 mm. Normalised distance d=0.80 (gate 4.0). No competing candidate within the gate.
Condition indicators	C1: latest depth 8.31 mm = 87% of nominal WT 9.5 mm (B23-0147, R2023) >= demo threshold 40%.
Evidence quality	Limited: E9: depth measurement uncertainty not stated; change cannot be tested against noise.
Engineering readiness	Inputs present - verification required. Missing: none. Verify: Depth measurement uncertainty (not stated); Tool sizing accuracy and any bias between inspection tools (always required)
System proposed next step	Send for engineering review
Human review	Not reviewed
Anomaly flags	Unusually deep relative to metadata; Unusual rate vs. population
Contradictions	none
Evidence gaps	Measurement uncertainty not stated; Unresolved measurement anomaly (High)
Identity resolution (reviewer)	none recorded

T0157 - Review first - Likely match

Evidence	Earlier (R2016, 2016-01-15)	Later (R2023, 2023-01-15)
Observation ID (source record)	A16-0126	B23-0144
OBSERVED axial position m	2050.637	2051.174
OBSERVED clock h	10.27	10.20
OBSERVED depth mm (1-sigma)	1.62 (0.40)	4.87 (0.30)
OBSERVED length / width mm	71 / 18	74 / 13
Nominal WT mm (design)	9.50	9.50
Record quality flags	-	-

Result	Value
CALCULATED observed change	3.25 mm over 7.00 yr
CALCULATED annualised change	0.464 mm/yr
UNCERTAINTY (measurement only)	sigma 0.071 mm/yr; 2-sigma range 0.321 to 0.607
Change vs noise	Increase exceeds measurement noise
Latest depth % WT	51.3
PREDICTED (illustrative)	Projection available (central + upper scenario): central 7.19 mm, upper 7.91 mm at +5 yr
Years to demo threshold (upper)	0.0
Forecast limitation	Constant-rate projection using the baseline observed rate; upper scenario adds 2 sigma of measurement noise. Only two observations: long-term trend is NOT established.

Basis	Detail
Match rationale	Aligned axial difference +0.13 m (raw +0.54 m, odometer offset +0.41 m), clock difference 0.07 h, length difference +3 mm. Normalised distance d=0.56 (gate 4.0). No competing candidate within the gate.
Condition indicators	C1: latest depth 4.87 mm = 51% of nominal WT 9.5 mm (B23-0144, R2023) >= demo threshold 40%. | C2: upper screening scenario (0.607 mm/yr) reaches the demo 50% WT threshold in 0.0 yr (<= 5-yr horizon). Illustrative, not a deadline. | C3: observed change +3.25 mm over 7.0 yr = +0.464 mm/yr (A16-0126 -> B23-0144) exceeds 2 sigma noise (0.50 mm) and >= demo rate 0.2 mm/yr.
Evidence quality	Adequate: No evidence issue detected
Engineering readiness	Inputs present for an engineer-selected assessment (not performed here). Missing: none. Verify: Tool sizing accuracy and any bias between inspection tools (always required)
System proposed next step	Send for engineering review
Human review	Not reviewed
Anomaly flags	Unusual rate vs. population; Change large relative to uncertainty
Contradictions	none
Evidence gaps	none
Identity resolution (reviewer)	none recorded

T0291 - Review first - Likely match

Evidence	Earlier (R2016, 2016-01-15)	Later (R2023, 2023-01-15)
Observation ID (source record)	A16-0219	B23-0270
OBSERVED axial position m	3823.050	3823.774
OBSERVED clock h	1.94	2.06
OBSERVED depth mm (1-sigma)	3.30 (0.40)	5.28 (0.30)
OBSERVED length / width mm	41 / 27	59 / 25
Nominal WT mm (design)	9.50	9.50
Record quality flags	-	-

Result	Value
CALCULATED observed change	1.98 mm over 7.00 yr
CALCULATED annualised change	0.283 mm/yr
UNCERTAINTY (measurement only)	sigma 0.071 mm/yr; 2-sigma range 0.140 to 0.426
Change vs noise	Increase exceeds measurement noise
Latest depth % WT	55.6
PREDICTED (illustrative)	Projection available (central + upper scenario): central 6.69 mm, upper 7.41 mm at +5 yr
Years to demo threshold (upper)	0.0
Forecast limitation	Constant-rate projection using the baseline observed rate; upper scenario adds 2 sigma of measurement noise. Only two observations: long-term trend is NOT established.

Basis	Detail
Match rationale	Aligned axial difference +0.21 m (raw +0.72 m, odometer offset +0.52 m), clock difference 0.12 h, length difference +18 mm. Normalised distance d=1.25 (gate 4.0). No competing candidate within the gate.
Condition indicators	C1: latest depth 5.28 mm = 56% of nominal WT 9.5 mm (B23-0270, R2023) >= demo threshold 40%. | C2: upper screening scenario (0.426 mm/yr) reaches the demo 50% WT threshold in 0.0 yr (<= 5-yr horizon). Illustrative, not a deadline. | C3: observed change +1.98 mm over 7.0 yr = +0.283 mm/yr (A16-0219 -> B23-0270) exceeds 2 sigma noise (0.50 mm) and >= demo rate 0.2 mm/yr.
Evidence quality	Adequate: No evidence issue detected
Engineering readiness	Inputs present for an engineer-selected assessment (not performed here). Missing: none. Verify: Tool sizing accuracy and any bias between inspection tools (always required)
System proposed next step	Send for engineering review
Human review	Not reviewed
Anomaly flags	Change large relative to uncertainty
Contradictions	none
Evidence gaps	none
Identity resolution (reviewer)	none recorded

T0342 - Review first - Likely match

Evidence	Earlier (R2016, 2016-01-15)	Later (R2023, 2023-01-15)
Observation ID (source record)	A16-0254	B23-0317
OBSERVED axial position m	4447.606	4448.403
OBSERVED clock h	7.74	8.00
OBSERVED depth mm (1-sigma)	0.92 (0.40)	3.41 (0.30)
OBSERVED length / width mm	18 / 8	20 / 18
Nominal WT mm (design)	7.90	7.90
Record quality flags	-	-

Result	Value
CALCULATED observed change	2.49 mm over 7.00 yr
CALCULATED annualised change	0.356 mm/yr
UNCERTAINTY (measurement only)	sigma 0.071 mm/yr; 2-sigma range 0.213 to 0.499
Change vs noise	Increase exceeds measurement noise
Latest depth % WT	43.2
PREDICTED (illustrative)	Projection available (central + upper scenario): central 5.19 mm, upper 5.90 mm at +5 yr
Years to demo threshold (upper)	1.1
Forecast limitation	Constant-rate projection using the baseline observed rate; upper scenario adds 2 sigma of measurement noise. Only two observations: long-term trend is NOT established.

Basis	Detail
Match rationale	Aligned axial difference -0.11 m (raw +0.80 m, odometer offset +0.91 m), clock difference 0.26 h, length difference +3 mm. Normalised distance d=0.79 (gate 4.0). No competing candidate within the gate.
Condition indicators	C1: latest depth 3.41 mm = 43% of nominal WT 7.9 mm (B23-0317, R2023) >= demo threshold 40%. | C2: upper screening scenario (0.499 mm/yr) reaches the demo 50% WT threshold in 1.1 yr (<= 5-yr horizon). Illustrative, not a deadline. | C3: observed change +2.49 mm over 7.0 yr = +0.356 mm/yr (A16-0254 -> B23-0317) exceeds 2 sigma noise (0.50 mm) and >= demo rate 0.2 mm/yr.
Evidence quality	Adequate: No evidence issue detected
Engineering readiness	Inputs present for an engineer-selected assessment (not performed here). Missing: none. Verify: Tool sizing accuracy and any bias between inspection tools (always required)
System proposed next step	Send for engineering review
Human review	Not reviewed
Anomaly flags	Change large relative to uncertainty
Contradictions	Priority hinges on a measurement within its own uncertainty of the threshold
Evidence gaps	none
Identity resolution (reviewer)	none recorded

T0352 - Review first - Likely match

Evidence	Earlier (R2016, 2016-01-15)	Later (R2023, 2023-01-15)
Observation ID (source record)	A16-0258	B23-0325
OBSERVED axial position m	4609.984	4610.705
OBSERVED clock h	5.74	5.36
OBSERVED depth mm (1-sigma)	2.72 (0.40)	3.86 (0.30)
OBSERVED length / width mm	12 / 44	17 / 43
Nominal WT mm (design)	7.90	7.90
Record quality flags	-	-

Result	Value
CALCULATED observed change	1.14 mm over 7.00 yr
CALCULATED annualised change	0.163 mm/yr
UNCERTAINTY (measurement only)	sigma 0.071 mm/yr; 2-sigma range 0.020 to 0.306
Change vs noise	Increase exceeds measurement noise
Latest depth % WT	48.9
PREDICTED (illustrative)	Projection available (central + upper scenario): central 4.67 mm, upper 5.39 mm at +5 yr
Years to demo threshold (upper)	0.3
Forecast limitation	Constant-rate projection using the baseline observed rate; upper scenario adds 2 sigma of measurement noise. Only two observations: long-term trend is NOT established.

Basis	Detail
Match rationale	Aligned axial difference -0.07 m (raw +0.72 m, odometer offset +0.79 m), clock difference 0.38 h, length difference +6 mm. Normalised distance d=1.03 (gate 4.0). No competing candidate within the gate.
Condition indicators	C1: latest depth 3.86 mm = 49% of nominal WT 7.9 mm (B23-0325, R2023) >= demo threshold 40%. | C2: upper screening scenario (0.306 mm/yr) reaches the demo 50% WT threshold in 0.3 yr (<= 5-yr horizon). Illustrative, not a deadline. | C4: observed change +1.14 mm over 7.0 yr = +0.163 mm/yr (A16-0258 -> B23-0325) exceeds 2 sigma noise (0.50 mm); below demo rate 0.2 mm/yr.
Evidence quality	Adequate: No evidence issue detected
Engineering readiness	Inputs present for an engineer-selected assessment (not performed here). Missing: none. Verify: Tool sizing accuracy and any bias between inspection tools (always required)
System proposed next step	Send for engineering review
Human review	Not reviewed
Anomaly flags	none
Contradictions	none
Evidence gaps	none
Identity resolution (reviewer)	none recorded

T0081 - Review first - Likely match

Evidence	Earlier (R2016, 2016-01-15)	Later (R2023, 2023-01-15)
Observation ID (source record)	A16-0067	B23-0077
OBSERVED axial position m	1160.746	1161.274
OBSERVED clock h	4.83	4.89
OBSERVED depth mm (1-sigma)	2.94 (0.40)	3.83 (0.30)
OBSERVED length / width mm	11 / 28	20 / 24
Nominal WT mm (design)	7.90	7.90
Record quality flags	-	-

Result	Value
CALCULATED observed change	0.89 mm over 7.00 yr
CALCULATED annualised change	0.127 mm/yr
UNCERTAINTY (measurement only)	sigma 0.071 mm/yr; 2-sigma range -0.016 to 0.270
Change vs noise	Change within measurement noise
Latest depth % WT	48.5
PREDICTED (illustrative)	Upper screening scenario only (growth not distinguishable from noise): central - mm, upper 5.18 mm at +5 yr
Years to demo threshold (upper)	0.4
Forecast limitation	Observed change is within measurement noise (|D| <= 2 sigma), so a central growth rate cannot be distinguished from zero. Only an upper screening scenario is shown.

Basis	Detail
Match rationale	Aligned axial difference +0.02 m (raw +0.53 m, odometer offset +0.51 m), clock difference 0.06 h, length difference +8 mm. Normalised distance d=0.46 (gate 4.0). Closest competing candidate A16-0068->B23-0077 at d=2.91 (margin 2.45 >= 1.00).
Condition indicators	C1: latest depth 3.83 mm = 48% of nominal WT 7.9 mm (B23-0077, R2023) >= demo threshold 40%. | C2: upper screening scenario (0.270 mm/yr) reaches the demo 50% WT threshold in 0.4 yr (<= 5-yr horizon). Illustrative, not a deadline.
Evidence quality	Adequate: No evidence issue detected
Engineering readiness	Inputs present for an engineer-selected assessment (not performed here). Missing: none. Verify: Tool sizing accuracy and any bias between inspection tools (always required)
System proposed next step	Send for engineering review
Human review	Not reviewed
Anomaly flags	none
Contradictions	Conclusion depends on the assumed growth rate
Evidence gaps	none
Identity resolution (reviewer)	none recorded

T0113 - Review first - Likely match

Evidence	Earlier (R2016, 2016-01-15)	Later (R2023, 2023-01-15)
Observation ID (source record)	A16-0087	B23-0102
OBSERVED axial position m	1524.058	1524.353
OBSERVED clock h	6.16	6.37
OBSERVED depth mm (1-sigma)	2.16 (0.40)	4.72 (0.30)
OBSERVED length / width mm	72 / 19	83 / 24
Nominal WT mm (design)	11.10	11.10
Record quality flags	-	-

Result	Value
CALCULATED observed change	2.56 mm over 7.00 yr
CALCULATED annualised change	0.366 mm/yr
UNCERTAINTY (measurement only)	sigma 0.071 mm/yr; 2-sigma range 0.223 to 0.509
Change vs noise	Increase exceeds measurement noise
Latest depth % WT	42.5
PREDICTED (illustrative)	Projection available (central + upper scenario): central 6.55 mm, upper 7.26 mm at +5 yr
Years to demo threshold (upper)	1.6
Forecast limitation	Constant-rate projection using the baseline observed rate; upper scenario adds 2 sigma of measurement noise. Only two observations: long-term trend is NOT established.

Basis	Detail
Match rationale	Aligned axial difference -0.11 m (raw +0.30 m, odometer offset +0.40 m), clock difference 0.21 h, length difference +11 mm. Normalised distance d=0.87 (gate 4.0). No competing candidate within the gate.
Condition indicators	C1: latest depth 4.72 mm = 43% of nominal WT 11.1 mm (B23-0102, R2023) >= demo threshold 40%. | C2: upper screening scenario (0.509 mm/yr) reaches the demo 50% WT threshold in 1.6 yr (<= 5-yr horizon). Illustrative, not a deadline. | C3: observed change +2.56 mm over 7.0 yr = +0.366 mm/yr (A16-0087 -> B23-0102) exceeds 2 sigma noise (0.50 mm) and >= demo rate 0.2 mm/yr.
Evidence quality	Adequate: No evidence issue detected
Engineering readiness	Inputs present for an engineer-selected assessment (not performed here). Missing: none. Verify: Tool sizing accuracy and any bias between inspection tools (always required)
System proposed next step	Send for engineering review
Human review	Not reviewed
Anomaly flags	Change large relative to uncertainty
Contradictions	Priority hinges on a measurement within its own uncertainty of the threshold
Evidence gaps	none
Identity resolution (reviewer)	none recorded

7. Candidate review plan (budget planner - illustrative)

Benefit points and costs are demo assumptions. The plan is a candidate for human decision; nothing is approved.

Finding	Category	Action	Assumed cost	Benefit points	Reason
T0153	Review first	Engineering review	1,500	30	Selected in the optimal set
T0177	Review first	Engineering review	1,500	30	Selected in the optimal set
T0158	Review first	Engineering review	1,500	30	Selected in the optimal set
T0160	Review first	Engineering review	1,500	30	Selected in the optimal set
T0157	Review first	Engineering review	1,500	30	Selected in the optimal set
T0291	Review first	Engineering review	1,500	30	Selected in the optimal set
T0342	Review first	Engineering review	1,500	30	Selected in the optimal set
T0352	Review first	Engineering review	1,500	30	Selected in the optimal set
T0081	Review first	Engineering review	1,500	30	Selected in the optimal set
T0113	Review first	Engineering review	1,500	30	Selected in the optimal set
T0162	Review first	Engineering review	1,500	30	Selected in the optimal set
T0094	Review first	Engineering review	1,500	30	Selected in the optimal set
T0307	Review first	Engineering review	1,500	30	Selected in the optimal set
T0003	Review first	Engineering review	1,500	30	Selected in the optimal set
T0207	Review first	Engineering review	1,500	30	Selected in the optimal set
T0066	Review first	Engineering review	1,500	30	Selected in the optimal set
T0142	Review first	Engineering review	1,500	30	Selected in the optimal set
T0252	Review first	Engineering review	1,500	30	Selected in the optimal set
T0337	Review first	Engineering review	1,500	30	Selected in the optimal set
T0076	Review first	Engineering review	1,500	30	Selected in the optimal set
T0021	Review first	Engineering review	1,500	30	Selected in the optimal set
T0186	Review first	Engineering review	1,500	30	Selected in the optimal set
T0134	Review first	Engineering review	1,500	30	Selected in the optimal set
T0018	Review first	Engineering review	1,500	30	Selected in the optimal set
T0073	Review first	Engineering review	1,500	30	Selected in the optimal set
T0127	Review first	Engineering review	1,500	30	Selected in the optimal set
T0356	Review first	Engineering review	1,500	30	Selected in the optimal set
T0143	Review first	Engineering review	1,500	30	Selected in the optimal set
T0278	Review first	Engineering review	1,500	30	Selected in the optimal set
T0203	Review first	Engineering review	1,500	30	Selected in the optimal set
T0184	Review first	Engineering review	1,500	30	Selected in the optimal set
T0163	Review first	Engineering review	1,500	30	Selected in the optimal set
T0221	Review first	Engineering review	1,500	30	Selected in the optimal set
T0152	Verify match / data	Match / sizing verification	600	9	Selected in the optimal set
T0151	Verify match / data	Match / sizing verification	600	9	Selected in the optimal set
T0082	Verify match / data	Match / sizing verification	600	9	Selected in the optimal set

Finding	Category	Action	Assumed cost	Benefit points	Reason
T0345	Verify match / data	Match / sizing verification	600	9	Selected in the optimal set
T0104	Verify match / data	Match / sizing verification	600	9	Selected in the optimal set
T0097	Verify match / data	Match / sizing verification	600	9	Selected in the optimal set
T0096	Verify match / data	Match / sizing verification	600	9	Selected in the optimal set
T0266	Verify match / data	Match / sizing verification	600	9	Selected in the optimal set
T0265	Verify match / data	Match / sizing verification	600	9	Selected in the optimal set
T0173	Verify match / data	Match / sizing verification	600	9	Selected in the optimal set
T0172	Verify match / data	Match / sizing verification	600	9	Selected in the optimal set
T0239	Verify match / data	Match / sizing verification	600	9	Selected in the optimal set
T0365	Verify match / data	Match / sizing verification	600	9	Selected in the optimal set
T0238	Verify match / data	Match / sizing verification	600	9	Selected in the optimal set
T0264	Verify match / data	Match / sizing verification	600	9	Selected in the optimal set
T0273	Verify match / data	Match / sizing verification	600	9	Selected in the optimal set
T0220	Verify match / data	Match / sizing verification	600	9	Selected in the optimal set
T0148	Additional data needed	Data request and follow-up	300	4	Selected in the optimal set

7b. Contradictions, anomalies, evidence gaps and reviewer proposals

Measurement anomaly flags: 43; contradictions: 41; evidence gaps: 226. Flags mark measurements to check; they are not corrosion events and no source value was changed.

Finding	Rule	Evidence A	Evidence B	Conflict	Would resolve
T0342	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 3.41 mm triggers the demo 40% WT rule (3.16 mm)	Measurement range +/-2 sigma: 2.81 to 4.01 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0081	K3 Conclusion depends on the assumed growth rate	Upper scenario reaches the demo 50% WT threshold in 0.4 yr	central projection not established	Whether the threshold is reached within the horizon depends on which growth rate is assumed.	A further inspection (third data point) or tighter sizing uncertainty
T0113	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 4.72 mm triggers the demo 40% WT rule (4.44 mm)	Measurement range +/-2 sigma: 4.12 to 5.32 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0094	K2 Large calculated change, but a source record has data-quality flags	Calculated change +1.82 mm (+0.260 mm/yr)	Data-quality flags on source records: INVALID_WALL_THICKNESS	The growth figure relies on records that validation questioned.	Confirm the flagged records with the inspection vendor
T0094	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 3.81 mm triggers the demo 40% WT rule (3.80 mm)	Measurement range +/-2 sigma: 3.21 to 4.41 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0307	K3 Conclusion depends on the assumed growth rate	Upper scenario reaches the demo 50% WT threshold in 1.4 yr	central projection not established	Whether the threshold is reached within the horizon depends on which growth rate is assumed.	A further inspection (third data point) or tighter sizing uncertainty
T0307	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 3.61 mm triggers the demo 40% WT rule (3.16 mm)	Measurement range +/-2 sigma: 3.01 to 4.21 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0003	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 2.76 mm does not trigger the demo 40% WT rule (3.16 mm)	Measurement range +/-2 sigma: 2.16 to 3.36 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0207	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 4.34 mm does not trigger the demo 40% WT rule (4.44 mm)	Measurement range +/-2 sigma: 3.74 to 4.94 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0066	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 3.92 mm does not trigger the demo 40% WT rule (4.44 mm)	Measurement range +/-2 sigma: 3.32 to 4.52 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0142	K3 Conclusion depends on the assumed growth rate	Upper scenario reaches the demo 50% WT threshold in 3.2 yr	central scenario: 5.1 yr	Whether the threshold is reached within the horizon depends on which growth rate is assumed.	A further inspection (third data point) or tighter sizing uncertainty
T0142	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 4.31 mm does not trigger the demo 40% WT rule (4.44 mm)	Measurement range +/-2 sigma: 3.71 to 4.91 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0252	K3 Conclusion depends on the assumed growth rate	Upper scenario reaches the demo 50% WT threshold in 2.9 yr	central projection not established	Whether the threshold is reached within the horizon depends on which growth rate is assumed.	A further inspection (third data point) or tighter sizing uncertainty
T0252	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 4.05 mm triggers the demo 40% WT rule (3.80 mm)	Measurement range +/-2 sigma: 3.45 to 4.65 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0337	K3 Conclusion depends on the assumed growth rate	Upper scenario reaches the demo 50% WT threshold in 3.4 yr	central projection not established	Whether the threshold is reached within the horizon depends on which growth rate is assumed.	A further inspection (third data point) or tighter sizing uncertainty

Finding	Rule	Evidence A	Evidence B	Conflict	Would resolve
T0337	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 4.04 mm triggers the demo 40% WT rule (3.80 mm)	Measurement range +/-2 sigma: 3.44 to 4.64 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0076	K3 Conclusion depends on the assumed growth rate	Upper scenario reaches the demo 50% WT threshold in 3.7 yr	central projection not established	Whether the threshold is reached within the horizon depends on which growth rate is assumed.	A further inspection (third data point) or tighter sizing uncertainty
T0076	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 4.55 mm triggers the demo 40% WT rule (4.44 mm)	Measurement range +/-2 sigma: 3.95 to 5.15 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0021	K3 Conclusion depends on the assumed growth rate	Upper scenario reaches the demo 50% WT threshold in 3.8 yr	central projection not established	Whether the threshold is reached within the horizon depends on which growth rate is assumed.	A further inspection (third data point) or tighter sizing uncertainty
T0021	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 4.60 mm triggers the demo 40% WT rule (4.44 mm)	Measurement range +/-2 sigma: 4.00 to 5.20 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0186	K3 Conclusion depends on the assumed growth rate	Upper scenario reaches the demo 50% WT threshold in 4.6 yr	central scenario: 6.5 yr	Whether the threshold is reached within the horizon depends on which growth rate is assumed.	A further inspection (third data point) or tighter sizing uncertainty
T0134	K3 Conclusion depends on the assumed growth rate	Upper scenario reaches the demo 50% WT threshold in 4.7 yr	central scenario: 8.3 yr	Whether the threshold is reached within the horizon depends on which growth rate is assumed.	A further inspection (third data point) or tighter sizing uncertainty
T0073	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 4.51 mm triggers the demo 40% WT rule (4.44 mm)	Measurement range +/-2 sigma: 3.91 to 5.11 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0356	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 3.31 mm triggers the demo 40% WT rule (3.16 mm)	Measurement range +/-2 sigma: 2.71 to 3.91 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0163	K1 High condition indicators rest on an ambiguous match	Condition: C3: observed change +1.47 mm over 7.0 yr = +0.210 mm/yr (A16-0132 -> B23-0151) exceeds 2 sigma noise (0.50 mm) and >= demo rate 0.2 mm/yr.	Identity: ambiguous pairing A16-0132 -> B23-0151; alternatives A16-0132->B23-0150 (d=1.71)	The concerning value may belong to a different physical defect.	Defect Identity Conflict review; raw ILI signal comparison
T0221	K1 High condition indicators rest on an ambiguous match	Condition: C3: observed change +1.41 mm over 7.0 yr = +0.201 mm/yr (A16-0174 -> B23-0203) exceeds 2 sigma noise (0.50 mm) and >= demo rate 0.2 mm/yr.	Identity: ambiguous pairing A16-0174 -> B23-0203; alternatives A16-0174->B23-0202 (d=1.28); A16-0173->B23-0203 (d=2.25)	The concerning value may belong to a different physical defect.	Defect Identity Conflict review; raw ILI signal comparison
T0082	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 2.96 mm does not trigger the demo 40% WT rule (3.16 mm)	Measurement range +/-2 sigma: 2.36 to 3.56 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0345	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 2.73 mm does not trigger the demo 40% WT rule (3.16 mm)	Measurement range +/-2 sigma: 2.13 to 3.33 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0104	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 2.70 mm does not trigger the demo 40% WT rule (3.16 mm)	Measurement range +/-2 sigma: 1.90 to 3.50 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0040	K5 Measured depth decreased, contradicting irreversible metal loss	A16-0034: 1.71 mm (earlier)	B23-0039: 0.47 mm (later)	Depth fell -1.24 mm, beyond 2-sigma noise (0.50 mm); metal loss does not reverse.	Check the match, tool-to-tool sizing bias, and whether the anomaly was repaired
T0106	K6 Deep defect with no earlier record	B23-0096: 2.34 mm = 30% WT in the later run	No earlier record within the matching gate	A defect this deep would usually have been reportable earlier unless it grew fast.	Earlier-run raw data near this location; check for a missed or mis-located call

Finding	Rule	Evidence A	Evidence B	Conflict	Would resolve
T0327	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 4.34 mm does not trigger the demo 40% WT rule (4.44 mm)	Measurement range +/-2 sigma: 3.74 to 4.94 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0243	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 3.33 mm does not trigger the demo 40% WT rule (3.80 mm)	Measurement range +/-2 sigma: 2.73 to 3.93 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0333	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 3.38 mm does not trigger the demo 40% WT rule (3.80 mm)	Measurement range +/-2 sigma: 2.78 to 3.98 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0083	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 2.96 mm does not trigger the demo 40% WT rule (3.16 mm)	Measurement range +/-2 sigma: 2.36 to 3.56 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0130	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 3.48 mm does not trigger the demo 40% WT rule (3.80 mm)	Measurement range +/-2 sigma: 2.88 to 4.08 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0193	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 2.73 mm does not trigger the demo 40% WT rule (3.16 mm)	Measurement range +/-2 sigma: 2.13 to 3.33 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0341	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 3.37 mm does not trigger the demo 40% WT rule (3.80 mm)	Measurement range +/-2 sigma: 2.77 to 3.97 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0037	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 4.02 mm does not trigger the demo 40% WT rule (4.44 mm)	Measurement range +/-2 sigma: 3.42 to 4.62 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly
T0129	K4 Priority hinges on a measurement within its own uncertainty of the threshold	Latest depth 3.25 mm does not trigger the demo 40% WT rule (3.80 mm)	Measurement range +/-2 sigma: 2.65 to 3.85 mm spans the threshold	The depth-based priority could flip within measurement uncertainty.	Higher-accuracy sizing (e.g. field verification) of this anomaly

Model Challenge Arena (held-out synthetic defects; not real-world accuracy)

Method	MAE	RMSE	Bias	2-sigma coverage	Lowest RMSE in splits
Global shrinkage (model B)	0.0427	0.0556	-0.0034	0.946	18/25
Segment shrinkage	0.0429	0.0562	-0.0042	0.946	7/25
Baseline observed rate	0.0583	0.0727	-0.0040	0.967	0/25
Pooled mean (reference)	0.0593	0.0853	-0.0002	0.956	0/25

8. Human review log

No human review decisions have been recorded for this dataset.

CorroSight records review status only. It does not approve repairs, authorise work orders or change operating conditions. A closed review is not a verified repair. This Evidence Passport records analysis and review evidence; it does not certify pipeline safety or regulatory compliance.

9. Scope, configuration and limitations

Question	Status in CorroSight	Explanation
Corrosion-growth estimation	Calculated by CorroSight	Observed change and annualised rate between two matched observations, with measurement-noise interval.
Time to a demo screening threshold	Illustrative scenario only	Constant-rate projection against a configurable demo threshold. Not a repair deadline or remaining life.
Remaining-strength assessment	NOT performed	Requires an applicable method chosen and applied by a qualified engineer, with verified inputs.
Probability of pipeline failure	NOT estimated	Would require validated models, material/operational data and engineering judgement.

Setting (demo assumption unless stated)	Value
pos_gate_m	1.0
clock_gate_h	1.0
pos_scale_m	0.25
clock_scale_h	0.4
length_scale_mm	20.0
ambiguity_margin	1.0
align_runs	True
near_duplicate_pos_m	0.05
near_duplicate_clock_h	0.25
near_duplicate_depth_mm	0.3
near_duplicate_length_mm	5.0
suspicious_rate_mm_per_yr	0.6
depth_review_pct_wt	40.0
rate_review_mm_per_yr	0.2
screening_threshold_pct_wt	50.0
forecast_horizon_years	5.0
noise_k	2.0
forecast_method	baseline

Limitations

- CorroSight is a hackathon prototype, not a certified pipeline-integrity system.
- Review priorities are rule-based recommendations for further human review, not risk classifications, failure probabilities or repair decisions.
- No remaining-strength (e.g. ASME B31G family) assessment is performed. The readiness checker only reports which inputs are available or missing.
- Growth estimates from two inspections cannot establish a long-term trend; constant-rate projections are illustrative scenarios, not predictions of future corrosion.

- Uncertainty intervals include only stated depth measurement uncertainty (assumed 1-sigma, independent, unbiased, Gaussian). Matching, sizing-bias and model-form uncertainty are not included.
- Time to a screening threshold is not a repair deadline.
- Demo results come from a synthetic, fictional dataset and do not demonstrate real-world accuracy.
- All consequential decisions require review and approval by qualified engineers. The application never approves repairs, work orders or changes to operating conditions.

Qualified reviewer name	Role	Signature	Date